Physics Bachelor projects

Quantum Metrology and Laser Applications group

LaserLaB VU

2026

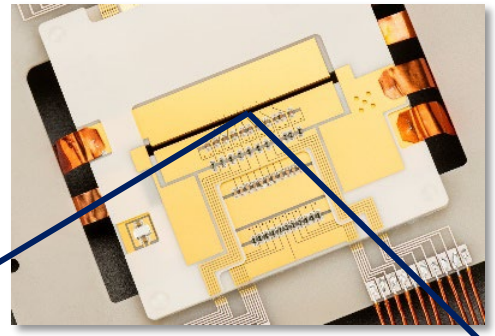
Quantum Metrology & Laser Applications

Overview of QMLA projects

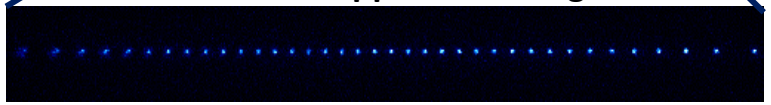
1. Structuring light for better optical clocks, qubits, and fundamental tests
2. Measuring a frequency chirp for laser cooling molecules
3. Programming a digital atomic clock Ethernet switch

1. Structuring light for better optical clocks, qubits, and fundamental tests

Picture of ion trap



Picture of trapped ion string

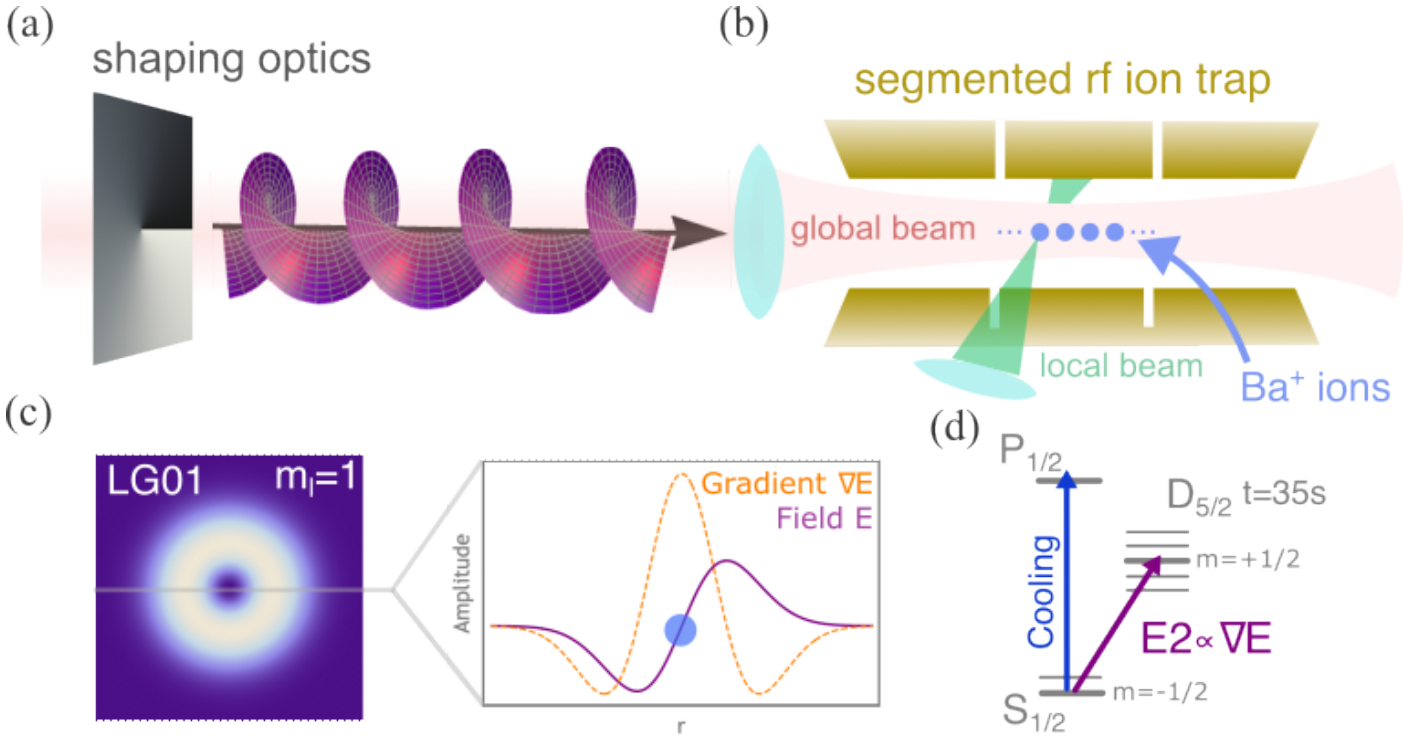


Are you interested in atomic physics and want to work on a state-of-the-art ion optical clock, join us and setup new techniques!

This project focuses on special exploring light to excite our atoms, even if they are completely in the dark! This helps with realizing better optical clocks, quantum qubit operations and tests at the frontier of of fundamental physics.

In this BSc project you will develop an experimental setup to shape beams into higher order Laguerre Gaussian (LG01) modes. You will characterize and optimize the beam. You will help implement it into the ion trap setup for actual spectroscopy.

More information: Laura Dreissen (l.s.dreissen@vu.nl)



2. Measuring a frequency chirp for laser cooling molecules

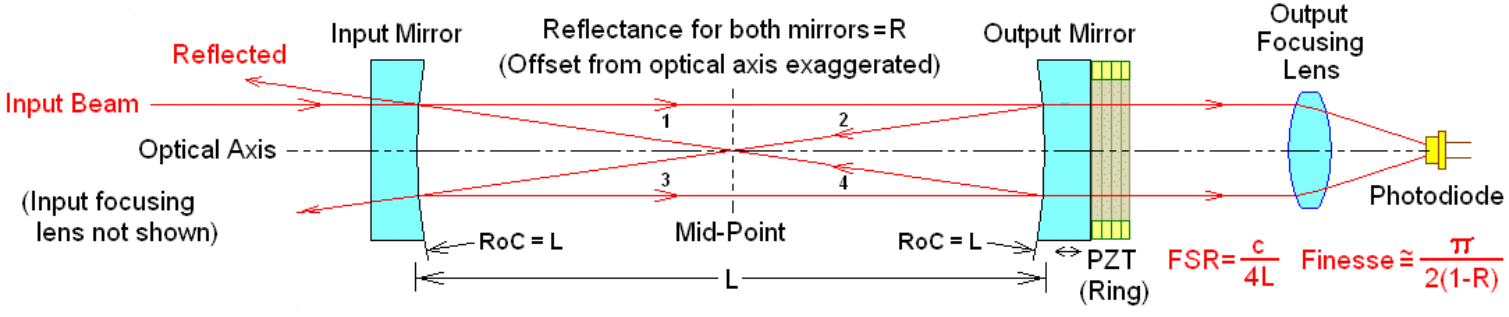
Are you excited about high-speed laser control, FPGA-based electronics, and laser cooling? This project focuses on building and characterizing an ultra-fast laser frequency-chirp system for laser cooling of He_2 molecules. We will use an AMD RFSoC platform to synthesize broadband, phase-continuous drive signals and apply serrodyne modulation with a fiber-coupled electro-optic modulator (EOM) to generate rapid, programmable frequency chirps. These chirps are needed to address multiple fine-structure components and to compensate the changing Doppler shift as molecules slow down during cooling.

A key part of the project is verifying the chirp with a homebuilt Fabry–Perot cavity. By recording the cavity transmission during a controlled scan, the chirped laser produces a characteristic distortion of the resonance peak compared to the unchirped case. From these measured traces (together with the cavity calibration), you can reconstruct the instantaneous frequency vs. time and benchmark chirp speed, linearity, and spurious sidebands.

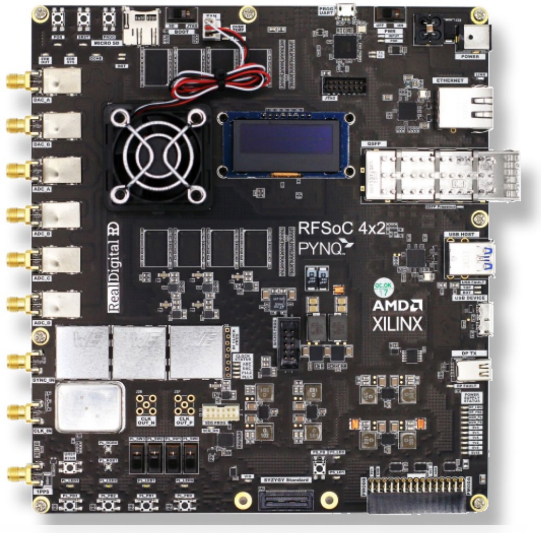
In this BSc project, you will:

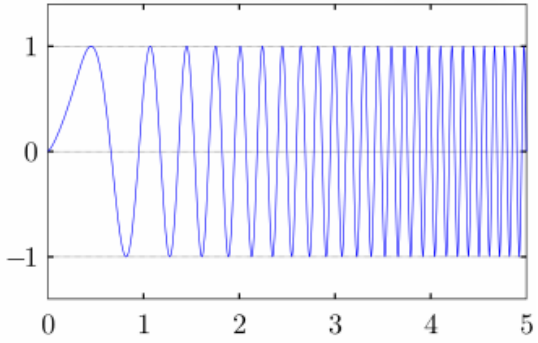
- Learn about RFSoC-based signal generation (FPGA + DAC), high-speed RF electronics, and waveform design for phase ramps.
- Implement serrodyne chirping with a fiber EOM and optimize for chirp efficiency and low spurs.
- Build an optical diagnostics setup with a scanning Fabry–Perot cavity and fast photodetection.
- Develop data acquisition + analysis to extract the frequency chirp.
- This project offers hands-on experience in FPGA/RF instrumentation, fiber/laser optics, precision cavity diagnostics, and experimental AMO physics—ideal for students passionate about building real hardware for fundamental measurements.

Interested? Learn more and pay a visit to our group or contact: Max Beyer (www.molecularions.com or m.beyer@vu.nl).



Confocal Cavity Scanning Fabry-Perot Interferometer

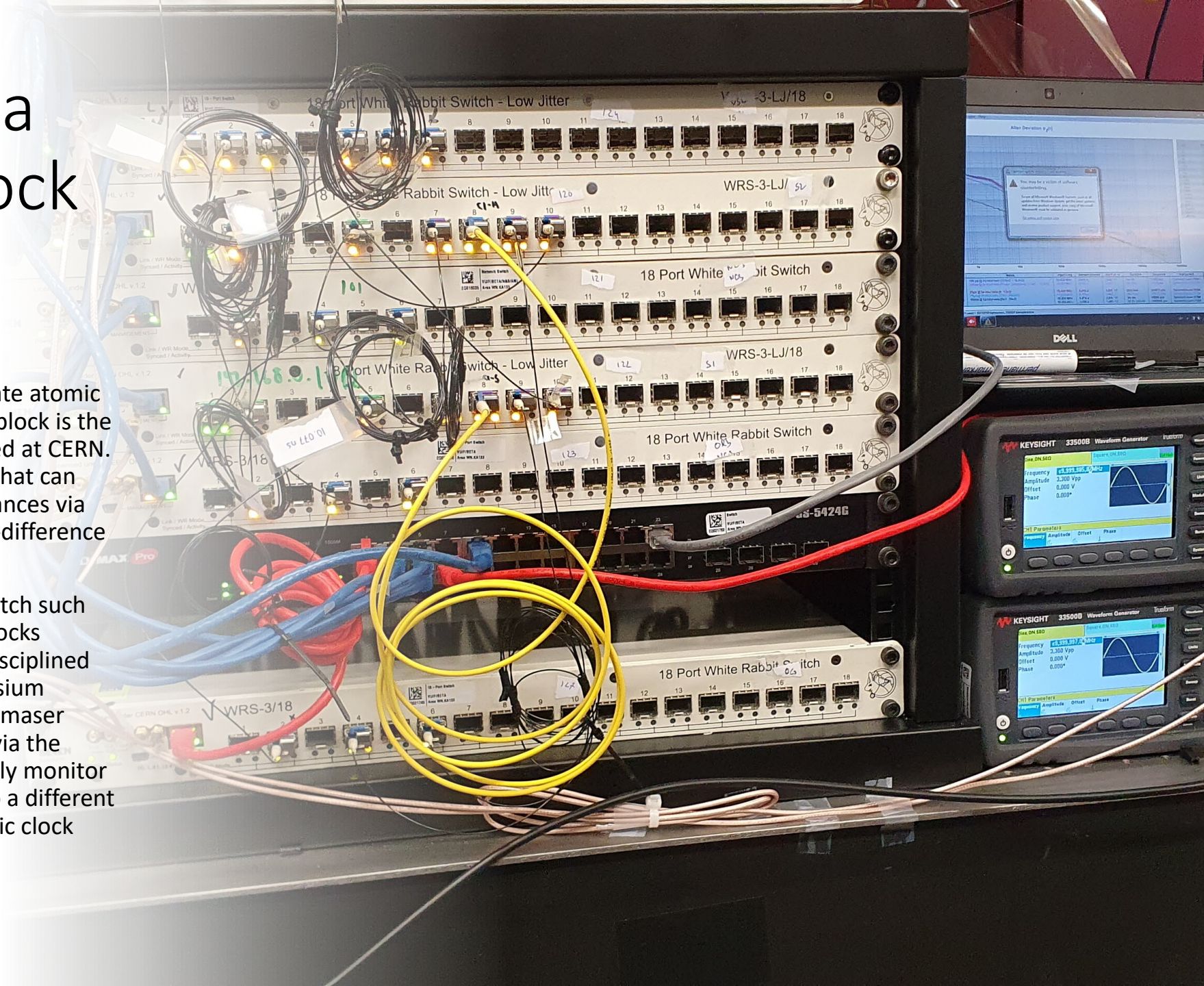




3. Programming a digital atomic clock Ethernet switch

PROGRAMMING SKILLS NEEDED

We are developing technology to fully integrate atomic clocks into telecom networks. A key building block is the White Rabbit (WR) Ethernet switch, developed at CERN. WR is a fully digital and open-source system that can transport atomic clock signals over large distances via optical fiber, and perform very precise phase-difference measurements.

In this BSc project you will program a WR switch such that it can 'listen' to three different atomic clocks signals: the time signal from a rooftop GPS-disciplined atomic clock, the signal from our in-house cesium atomic clock, and the signal from a hydrogen maser atomic clock at VSL Delft, transported to VU via the SURF network. This switch should continuously monitor the clock time differences, and switch over to a different atomic clock source in case the primary atomic clock source fails.

More information: Jeroen Koelemeij (j.c.j.koelemeij@vu.nl)